



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2012
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 66

- Attempt questions 1 – 6.
- All questions are not of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *P.R.Bigelow*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, x > 0$

Section A (20 Marks)

START A NEW BOOKLET

Question 1. (5 marks)

Indicate which of the answers A, B, C, or D is the correct answer. Write the answer in your answer booklet. **Marks**

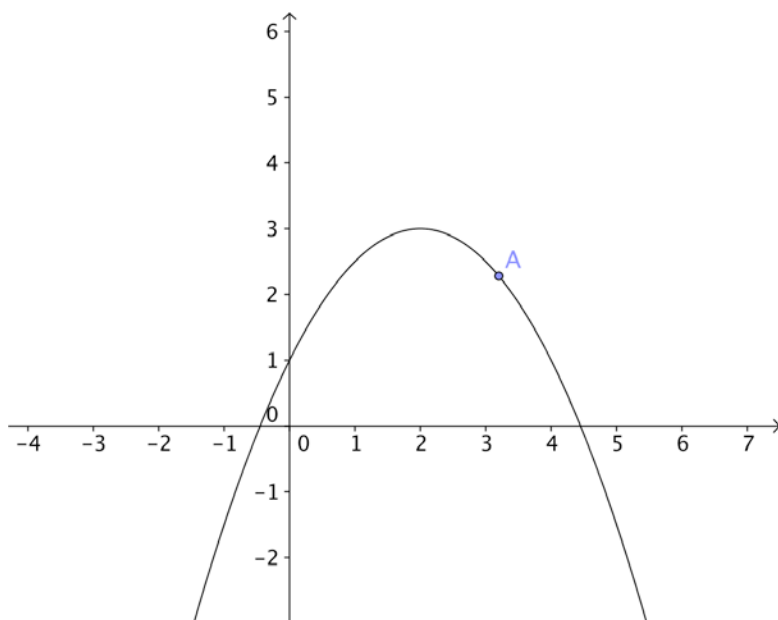
(a) $\frac{d}{dx}(\cos 3x)$ equals :- **A:** $\frac{1}{3}\sin 3x$ **1**

B: $3\sin 3x$

C: $-\frac{1}{3}\sin 3x$

D: $-3\sin 3x$

(b) **1**



The diagram shows $y = f(x)$. At point A:

A: $y' > 0, y'' > 0$

B: $y' < 0, y'' < 0$

C: $y' > 0, y'' < 0$

D: $y' < 0, y'' > 0$

(c) $4 \ln \sqrt{e}$ equals:- **A:** 2 **1**

B: $2e$

C: 1

D: $\sqrt{2}$

(d) $\int_{-3}^3 x^3 dx$ equals:- **1**

A: $\left| \int_{-3}^0 x^3 dx \right| + \int_0^3 x^3 dx$

B: $2 \int_0^3 x^3 dx$

C: 0

D: $\frac{81}{2}$

(e) If $f(x) = \sin x$, then $f'\left(-\frac{\pi}{3}\right)$ is:- **1**

A: $\frac{\sqrt{3}}{2}$

B: $\frac{1}{2}$

C: $-\frac{1}{2}$

D: $-\frac{\sqrt{3}}{2}$

Question 2 (15 marks)

Marks

5

(a) Differentiate the following:

(i) $y = \cos 4x$

(ii) $y = e^{-4x}$

(iii) $f(x) = x \tan x$

(iv) $f(x) = \frac{\sin x}{x}$

(v) $y = \cos^3 2x$

(b) Find

3

(i) $\int \frac{dx}{x+3}$

(ii) $\int_0^1 \frac{3}{e^{2x}}$

(c) Use Simpson's Rule with three function values to find an approximation to

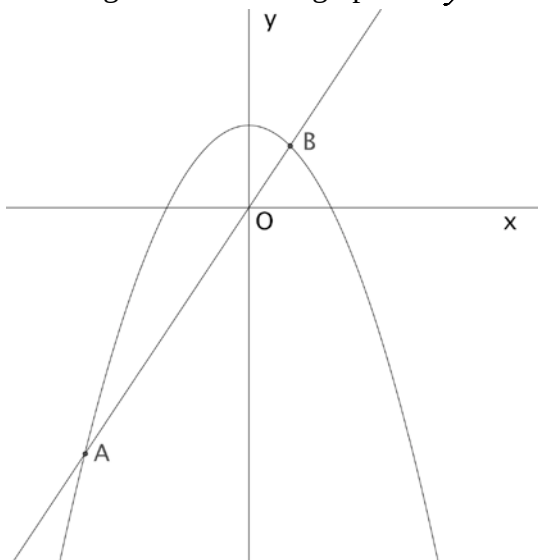
3

$$\int_0^1 \sqrt{1+x^3} dx$$

(Answer correct to two decimal places.)

(d) The diagram shows the graphs of $y = 4 - x^2$ and $y = 3x$.

4



(i) Find the x-values of the points of intersection A and B.

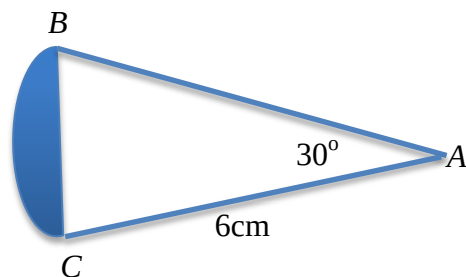
(ii) Calculate the area of the region between the two graphs.

Section B (20 Marks)

START A NEW BOOKLET

Question 3 (10 Marks)

- | | Marks |
|---|--------------|
| (a) Given $f(x) = 3\ln(2+x)$ find $f'(4)$. | 2 |
| (b) Find $\int_0^{\frac{\pi}{4}} \cos 2x \, dx$. | 2 |
| (c) Find the value of $\sqrt{e^3}$ correct to three significant figures. | 1 |
| (d) Use the identity $\sec^2 x = 1 + \tan^2 x$ to evaluate $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx$. | 2 |
| (e) In the diagram $\angle BAC = 30^\circ$ and a circular arc of radius 6 cm, centre A, is constructed from B to C. | 3 |



- (i) Find the area of $\triangle ABC$.
- (ii) Calculate the exact area of the shaded segment.

Question 4 (10 Marks)

(a) Find $f'(x)$ and $f''(x)$ where $f(x) = \sin(x^2)$. 3

(b) Given the function $y = x^3 - 9x + 3$ 7

- (i) Find the co-ordinates of the stationary points, and determine their nature.
- (ii) Find the co-ordinates of any points of inflexion.
- (iii) Sketch the curve in the domain $-4 \leq x \leq 4$.
- (iv) What is the greatest value of $x^3 - 9x + 3$ in the domain $-4 \leq x \leq 4$?

Section C (26 Marks)

START A NEW BOOKLET

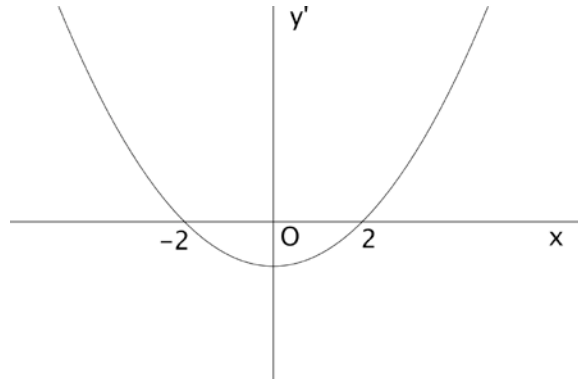
Question 5 (10 Marks)

- (a) For the function $f(x) = 2\cos x + 1$ 5
- (i) State the range of $f(x)$.
 - (ii) Sketch $f(x) = 2\cos x + 1$ for $0 \leq x \leq 2\pi$.
 - (iii) Calculate the exact area of the region in the first quadrant bounded by the curve $f(x) = 2\cos x + 1$, the y-axis, and the line $y = 1$.
- (b) 5
- (i) Sketch $y = \log_e x$. Mark on the curve the point P where $x = e$.
 - (ii) Find, in general form, the equation of the tangent to the curve $y = \log_e x$ at the point P .
 - (iii) Using the sketch of $y = \log_e x$ find the values of k for which $kx = \log_e x$ has at least one real root.

Question 6 (16 Marks)

- (a) Simplify $e^{2\ln 5}$. 1

- (b) 4



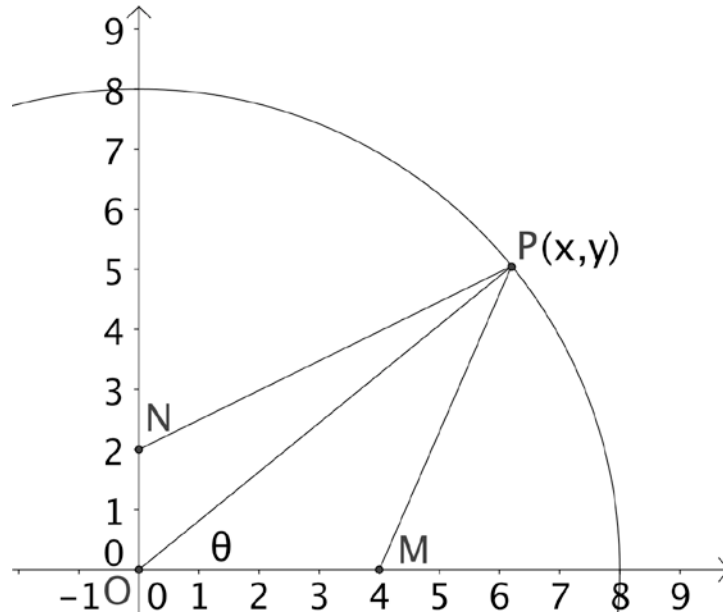
The diagram shows the graph of the gradient function of the curve $y = f(x)$.

- (i) What type of point occurs on $y = f(x)$ at $x = -2$?
Justify your answer.
- (ii) If $f(-2) = 5$ and $f(2) > 0$ sketch $y = f(x)$.
- (c) On the same (new) diagram: 4
- (i) Sketch $y = x$ and $y = x^2$ in the first quadrant.
- (ii) Mark the points of intersection and indicate, by shading, the area between the two graphs.
- (iii) Find the volume generated when this area is rotated about the x-axis.

(Question continued overleaf)

(d)

7



The diagram shows the part of the circle $x^2 + y^2 = 64$ in the first quadrant. The point $P(x,y)$ lies on the circle, centre O .

M is on the x -axis at $x = 4$, N is on the y -axis at $y = 2$, and $\angle MOP = \theta$ in radians.

- (i) Show that the area, A , of the quadrilateral $OMPN$ is given by

$$A = 16 \sin \theta + 8 \cos \theta$$

- (ii) Find the value of $\tan \theta$ for which A is a maximum.
- (iii) Hence find in surd form the co-ordinates of P for which A is a maximum.

This is the end of the paper.

SECTION A

1. a) D
b) B
c) A
d) C
e) B. ✓

2. a) i) $-4 \sin 4x$ ✓

ii) $-4e^{-4x}$ ✓

iii) $\tan x + x \sec^2 x$ ✓

iv) $\frac{x \cos x - \sin x}{x^2}$ ✓

v) $3(\cos 2x)^2(-2 \sin 2x)$
 $= -6 \cos^2 2x \sin 2x$ ✓

b) i) $\ln(x+3) + C$ ✓

ii) $3 \int_0^1 e^{-2x} \cdot dx = 3 \left[-\frac{1}{2} e^{-2x} \right]_0^1$
 $= -\frac{3}{2} e^{-2} - \left(-\frac{3}{2} e^0 \right)$
 $= \frac{3}{2} \left(1 - \frac{1}{e^2} \right)$

c) $h = b - a = \frac{1-0}{2} = \frac{1}{2}$

x	0	0.5	1
	1	$\sqrt{1.25}$	$\sqrt{2}$

$\int_0^1 \sqrt{1+x^3} \cdot dx = \int_a^b f(x) \cdot dx$

$\frac{h}{3} (y_0 + 4y_1 + 4y_3 + \dots + y_n)$
 $\frac{1}{6} (1 + 4\sqrt{2} + 4(\sqrt{1.25}) + 1)$
 $= 1.10947 \approx 1.11$ (2dp)

d) i) $y = 4 - x^2 = 3x$

$x^2 + 3x - 4 = 0$
 $(x-1)(x+4) = 0$

P.O.I $B(1, 3) A(-4, -12)$

ii) $\int_{-4}^1 4 - x^2 - 3x \cdot dx$
 $= \left[4x - \frac{x^3}{3} - \frac{3}{2}x^2 \right]_{-4}^1$
 $= \left[4 - \frac{1}{3} - \frac{3}{2} \right] - \left[-16 + \frac{64}{3} - \frac{48}{2} \right]$
 $= \frac{2}{6} - \left(-18 \frac{2}{3} \right)$
 $= \frac{20}{3} \text{ units}^2$

$$3 \text{ (a)} \quad f(x) = 3 \ln(2+x)$$

$$f'(x) = \frac{3}{2+x}$$

$$f'(4) = \frac{3}{2+4} = \frac{1}{2}$$

$$\begin{aligned} (b) \quad \int_0^{\frac{\pi}{4}} \cos 2x \cdot dx &= \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} (\sin \frac{\pi}{2} - \sin 0) \\ &= \frac{1}{2} \end{aligned}$$

$$(c) \quad \sqrt{e^3} = 4.48 \quad (3 \text{ sig. fig.})$$

$$\begin{aligned} (d) \quad \int_0^{\frac{\pi}{4}} \tan^2 x \cdot dx &= \int_0^{\frac{\pi}{4}} \sec^2 x - 1 \cdot dx \\ &= [\tan x - x]_0^{\frac{\pi}{4}} \\ &= (1 - \frac{\pi}{4}) - (0 - 0) \\ &= 1 - \frac{\pi}{4} \end{aligned}$$

$$(e) (i) \text{ Area } \Delta ABC = \frac{1}{2} \cdot 6^2 \cdot \sin 30 = 9 \text{ cm}^2$$

$$\begin{aligned} (ii) \text{ Area of shaded} &= \frac{1}{2} r^2 \theta - 9 \\ &= \frac{1}{2} \cdot 6^2 \cdot \frac{\pi}{6} - 9 \\ &= 3\pi - 9 \end{aligned}$$

$$4(a) \quad f(x) = \sin(x^2) \quad (\text{Chain Rule})$$

$$f'(x) = 2x \cdot \cos(x^2)$$

$$\begin{aligned} f''(x) &= 2x \cdot (-2x \sin(x^2)) \\ &\quad + \cos(x^2) \cdot 2 \quad (\text{Prod. Rule}) \\ &= 2 \cos(x^2) - 4x^2 \sin(x^2) \end{aligned}$$

$$(b) \quad y = x^3 - 9x + 3$$

$$y' = 3x^2 - 9$$

$$\text{For stat pts } y' = 0$$

$$\therefore 3(x^2 - 3) = 0$$

$$x = \pm \sqrt{3}$$

$$\text{Test in } y'' = 6x$$

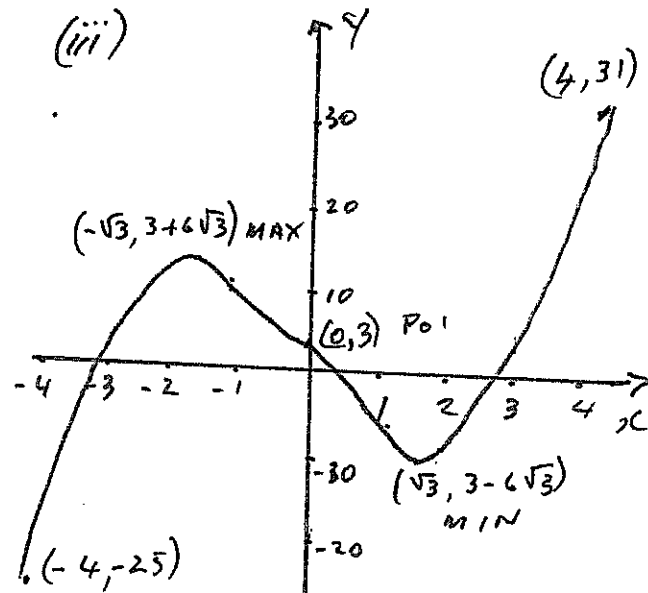
$$\text{at } x = \sqrt{3}; y'' > 0 \therefore \text{min}$$

$$\therefore (\sqrt{3}, 3 - 6\sqrt{3}) \text{ MIN}$$

$$\text{Similarly } (-\sqrt{3}, 3 + 6\sqrt{3}) \text{ MAX.}$$

$$\text{P.O.I at } x=0 \text{ and concavity changes. } (0, 3)$$

(iii)



(iv) Clearly max value of function in given domain is 31 at $x = 4$

SECTION C

Question 5

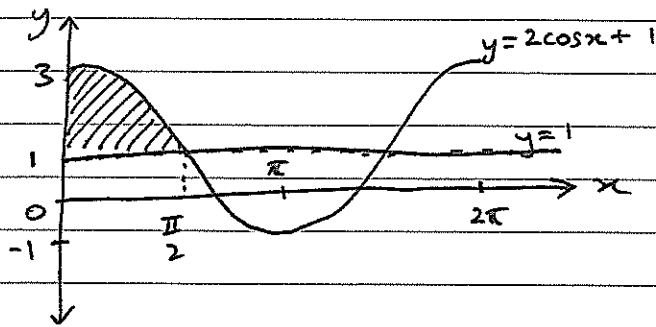
(a) i) $-1 \leq \cos x \leq 1$
 $-2 \leq 2\cos x \leq 2$
 $-1 \leq 2\cos x + 1 \leq 3$

R: $-1 \leq y \leq 3$

ii) $f(x) = 2\cos x + 1$ for $0 \leq x \leq 2\pi$

$a = 2$

$p = \frac{2\pi}{1}$



iii) $A = \int_0^{\frac{\pi}{2}} (2\cos x + 1 - 1) dx$

$$= \int_0^{\frac{\pi}{2}} 2\cos x dx$$

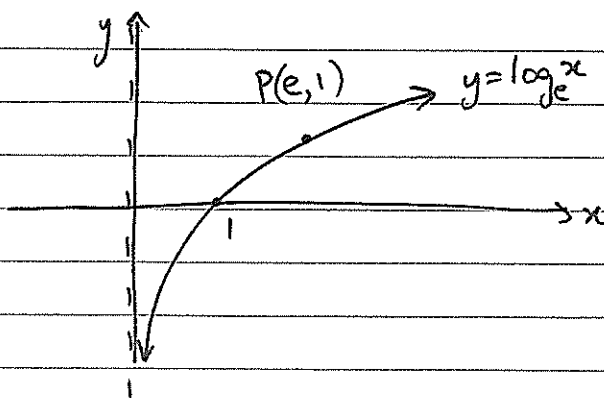
$$= \left[2\sin x \right]_0^{\frac{\pi}{2}}$$

$$= 2\sin \frac{\pi}{2} - 2\sin 0$$

$$= 2(1)$$

$$= 2 \text{ units}^2$$

(b) i)



ii) $y = \log_e x$

$$y' = \frac{1}{x}$$

at $P(e, 1)$

$$m_T = \frac{1}{e}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{e}(x - e)$$

$$ey - e = x - e$$

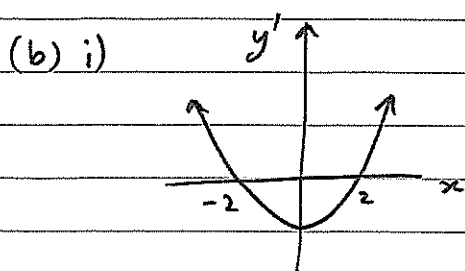
$$x - ey = 0$$

iii) The tangent at P found in (ii) passes through the origin since it's in the form $y = kx$ where $k = \frac{1}{e}$.

Answer is $-\infty < k \leq \frac{1}{e}$

Question 6

(a) $e^{2 \ln 5} = e^{\ln 5^2}$
 $= 5^2$
 $= 25$



when $x = -2$

$$y' = 0 \quad (y' \text{ cuts the } x\text{-axis})$$

$$y'' < 0 \quad (\text{gradient of } y' \text{ is negative})$$

↪ concave down

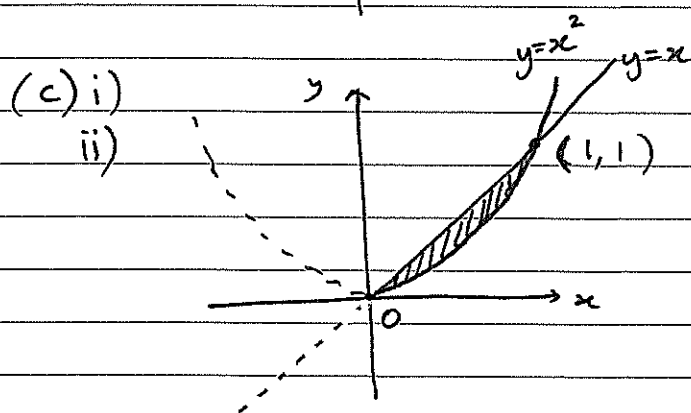
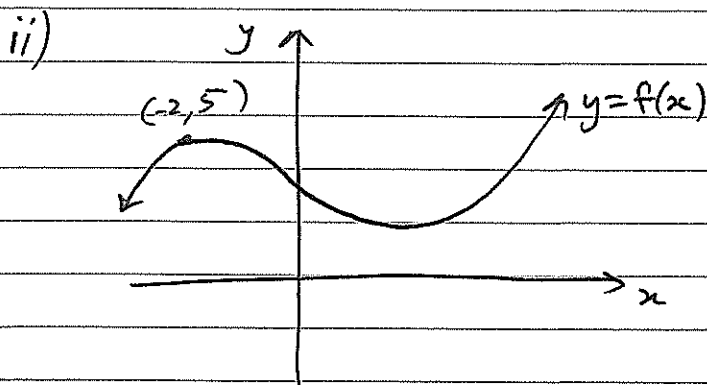
∴ Maximum Turning Point when $x = -2$

OR consider y' on either side of $x = -2$

x	-3	-2	-1
y'	+	0	-

$\leftarrow$ since y' is below x -axis
 $\uparrow$ since y' is above x -axis

$\therefore$ Maximum Turning Point when $x = -2$



$$\begin{aligned}
 \text{iii)} \quad V &= \pi \int_a^b y^2 dx \\
 &= \pi \int_0^1 ((x)^2 - (x^2)^2) dx \\
 &= \pi \int_0^1 (x^2 - x^4) dx \\
 &= \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 \\
 &= \pi \left[\frac{(1)^3}{3} - \frac{(1)^5}{5} - (0) \right] \\
 &= \frac{2\pi}{15} \text{ units}^3
 \end{aligned}$$

(d) assuming P lies in the first quadrant
 θ is acute.

$$\begin{aligned} \text{i) Area } \triangle OPM &= \frac{1}{2}(8)(4)\sin\theta \\ &= 16\sin\theta \end{aligned}$$

$$\begin{aligned} \text{Area } \triangle OPN &= \frac{1}{2}(2)(8)\sin\left(\frac{\pi}{2} - \theta\right) \\ &= 8\sin\left(\frac{\pi}{2} - \theta\right) \\ &= 8\cos\theta \quad (\text{complementary angles}) \end{aligned}$$

$$\text{Area of quadrilateral } OMPN = 16\sin\theta + 8\cos\theta$$

$$\text{ii) } A = 16\sin\theta + 8\cos\theta$$

$$\frac{dA}{d\theta} = 16\cos\theta - 8\sin\theta$$

$$\frac{d^2A}{d\theta^2} = -16\sin\theta - 8\cos\theta$$

$$\text{let } \frac{dA}{d\theta} = 0 \text{ for stat. points}$$

$$16\cos\theta - 8\sin\theta = 0$$

$$16\cos\theta = 8\sin\theta \quad (\div \cos\theta)$$

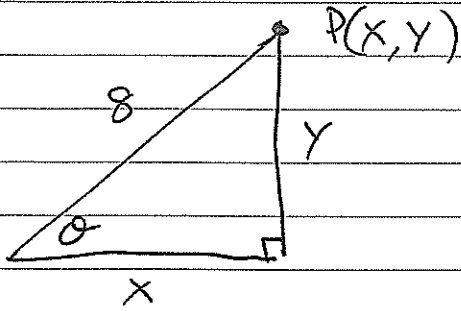
$$16 = 8\tan\theta$$

$$\tan\theta = 2$$

$$\text{when } \theta \text{ is acute } \frac{d^2A}{d\theta^2} < 0 \quad \curvearrowright \quad \text{concave down}$$

$\therefore A$ is a maximum when $\tan\theta = 2$

iii)

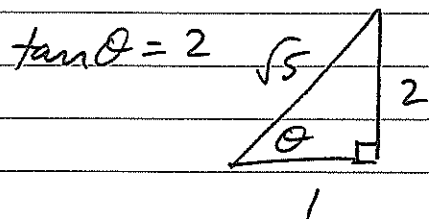


$$\cos \theta = \frac{x}{8}$$

$$\sin \theta = \frac{y}{8}$$

$$x = 8 \cos \theta$$

$$y = 8 \sin \theta$$



$$\cos \theta = \frac{1}{\sqrt{5}}$$

$$\sin \theta = \frac{2}{\sqrt{5}}$$

$$x = 8 \left(\frac{1}{\sqrt{5}} \right)$$

$$y = 8 \left(\frac{2}{\sqrt{5}} \right)$$

$$= \frac{8}{\sqrt{5}}$$

$$= \frac{16}{\sqrt{5}}$$

$$= \frac{8\sqrt{5}}{5}$$

$$= \frac{16\sqrt{5}}{5}$$

$\therefore P$ has coordinates $\left(\frac{8\sqrt{5}}{5}, \frac{16\sqrt{5}}{5} \right)$